

PACE TECHNOLOGIES

Metallographic Handbook



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11.6.2 Titanium and Titanium Alloys (Conventional Polishing)

Description:

Titanium alloys are very useful because they have a good strength-to-weight ratio. This makes them ideal for use in areas ranging from aerospace to sports equipment.

Preparation Challenge:

Metallographic preparation of titanium and titanium alloys is fairly straight forward by conventional metallographic techniques.

SECTIONING

MAXCUT abrasive blade (MAX-C or MAX-I series)

MOUNTING

Phenolic, epoxy or diallyl phthalate compression mounting resins

POLISHING

Abrasive/Surface	Lubrication	Force/ sample	Speed (Head/Base)	Time
240 (P220) grit SiC paper 360 (P500) grit SiC paper 600 (P1200) grit SiC paper 800 (P2400) grit SiC paper 1200 (P4000) grit SiC paper	Water	5-10 lbs	200/200 rpm	Until plane 1 minute 1 minute 1 minute 1 minute
6 micron DIAMAT diamond on TEXPAN polishing pad	DIALUBE Purple Extender	5-10 lbs	200/200 rpm	3 minutes
1 micron DIAMAT diamond on GOLDPAD polishing pad	DIALUBE Purple Extender	5-10 lbs	200/200 rpm	2 minutes
0.05 micron Nanometer alumina on ATLANTIS polishing pad		5-10 lbs	100/100 rpm	1 minute



Figure 11-47 Ti₆Al₄V Titanium Alloy, 400X (DIC), Etchant Kroll's Reagent.

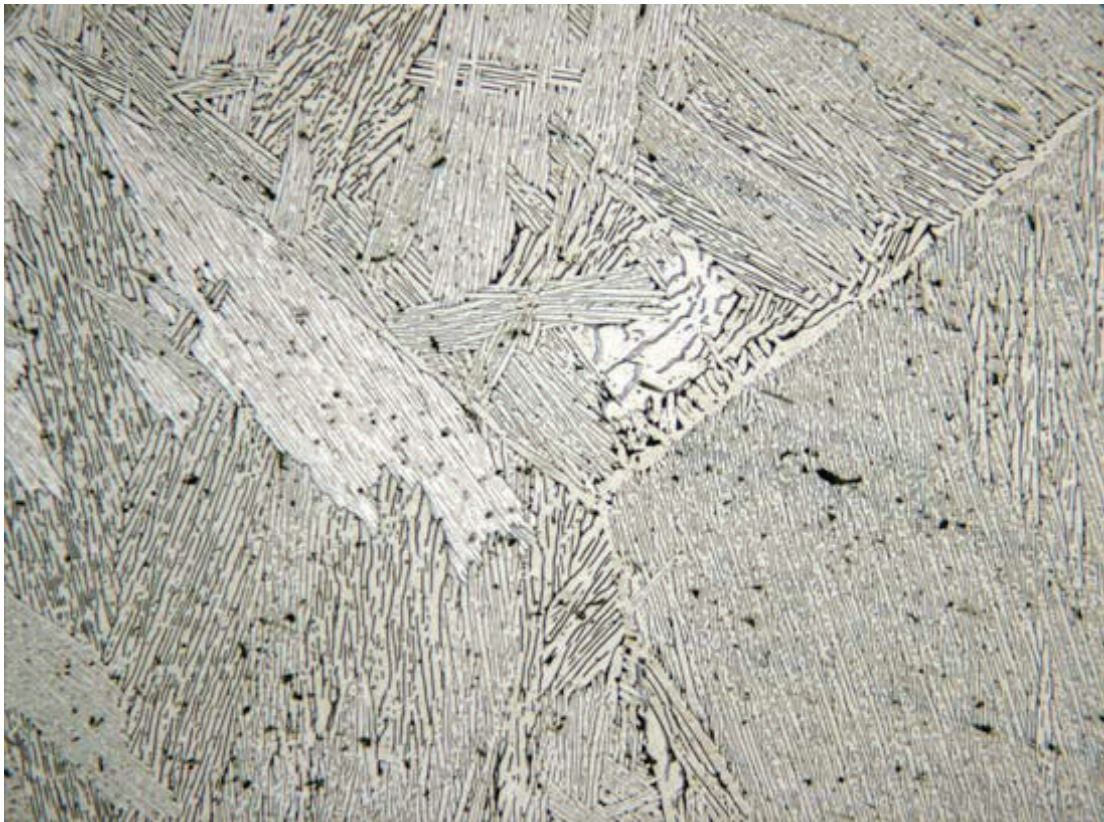


Figure 11-48 Cast Titanium, 200X (BF), Etchant Kroll's Reagent.

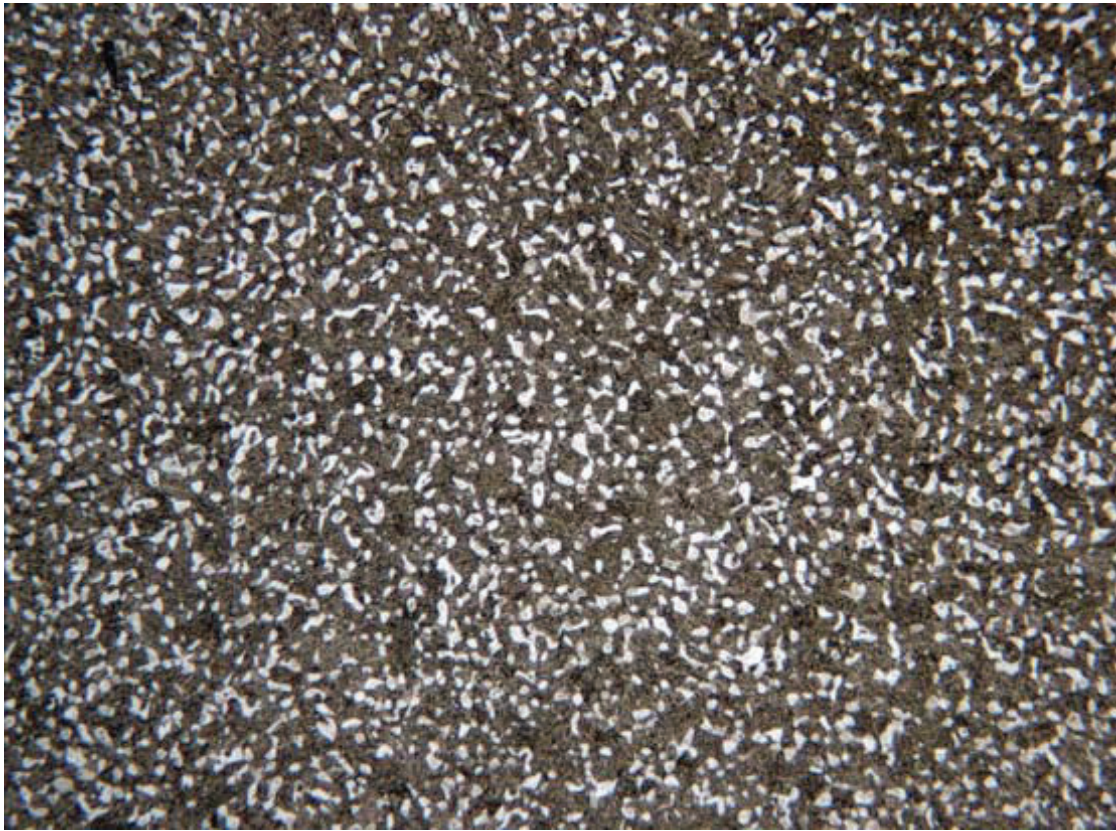


Figure 11-49 Wrought Titanium, 200X (BF), Etchant Ammonium Bifluoride.

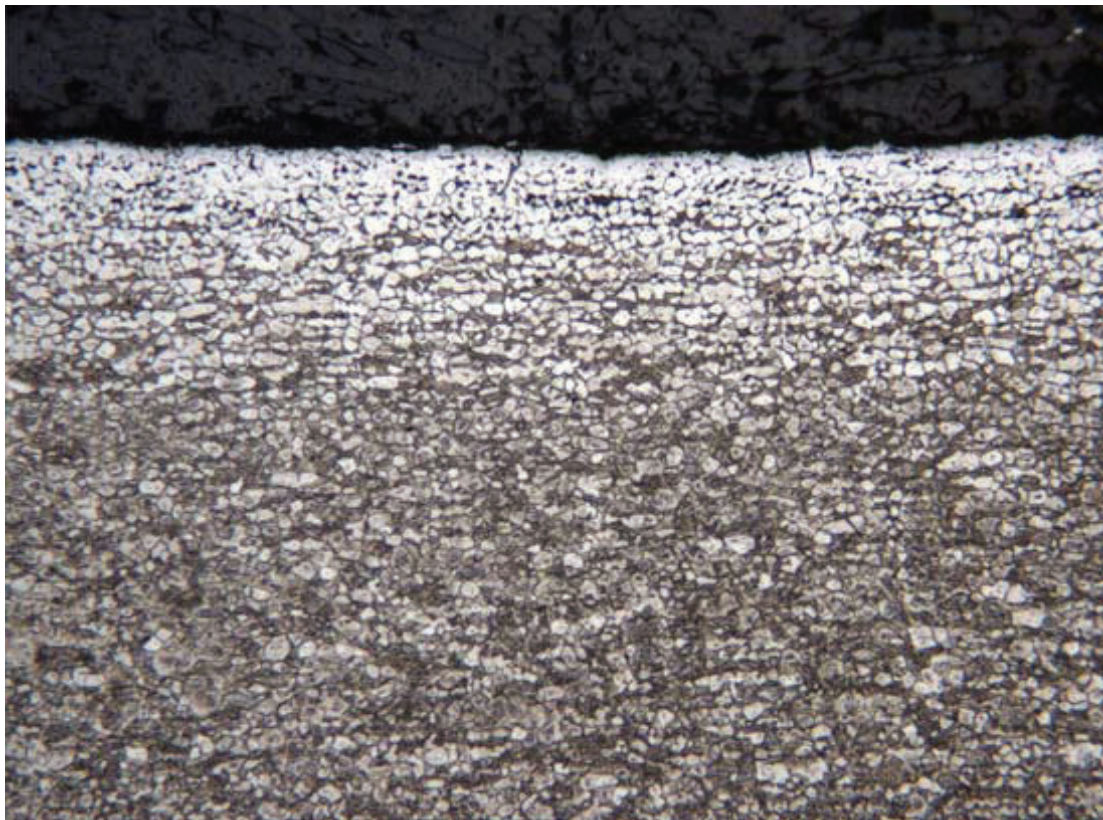


Figure 11-50 Alpha-rich Case Wrought Titanium, 200X (BF).

11.6.3 Titanium Alloy - Attack Polishing

(contributed by Jim Hallquist Medtronic Inc.)

Description:

Titanium is a very wear and corrosion resistant material, thus it has found some useful applications in the medical industry ranging from hip implants to pacemaker casings.

Preparation Challenge:

Specimen preparation of titanium is relatively straight forward; however, attack polishing has been successfully used to enhance microstructural features.

SECTIONING

MAXCUT abrasive blade (MAX-C or MAX-I series)

MOUNTING

Phenolic, epoxy or diallyl phthalate compression mounting resins

POLISHING

Abrasive/Surface	Lubrication	Force/ sample	Speed (Head/Base)	Time
180 grit SiC paper 240 (P220) grit SiC paper 320 (P360) grit SiC paper	water	5-10 lbs	200/200 rpm	Until plane 1 minute 1 minute
9 micron DIAMAT diamond on TEXPAN polishing pad		5-10 lbs	200/200 rpm	10 minutes
100 ml SIAMAT colloidal silica with 20 ml H ₂ O ₂ and 10 ml Kroll's reagent		5-10 lbs	100/100 rpm	10 minutes

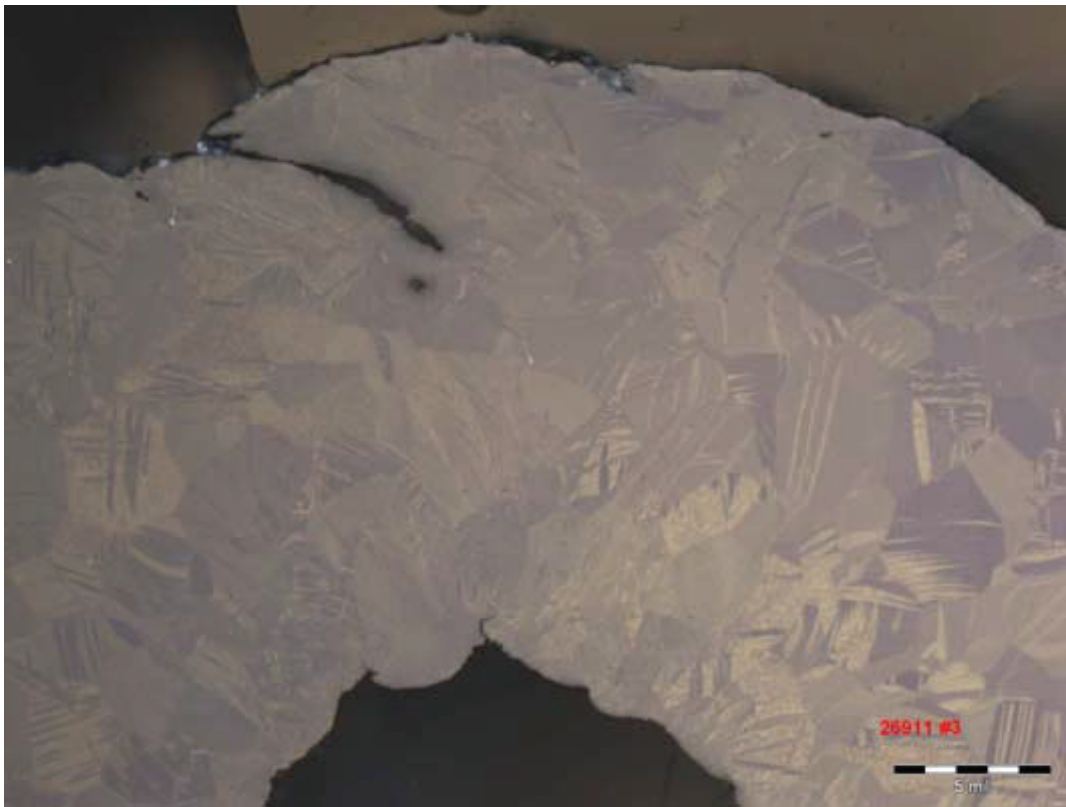


Figure 11-51 Titanium alloy, 500X, as attack polished with H_2O_2 /Kroll's reagent (photo courtesy of Medtronic Inc).

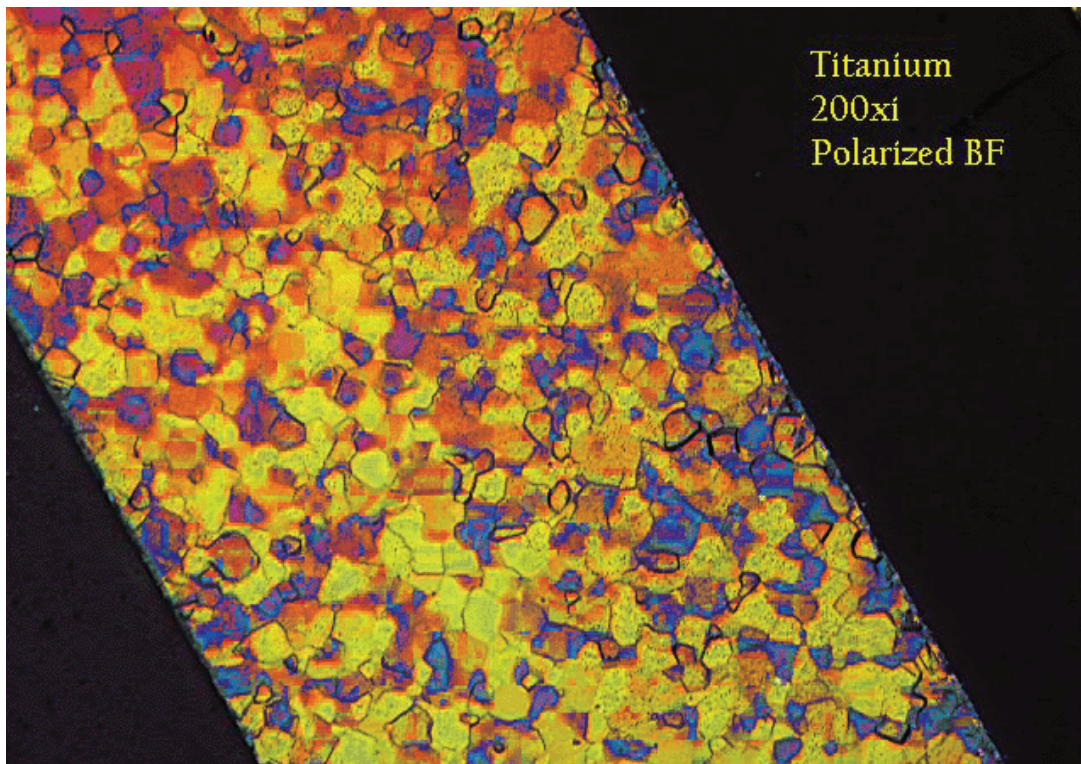


Figure 11-52 Titanium alloy, 200X (polarized light), as attack polished with H_2O_2 /Kroll's reagent (photo courtesy of Medtronic Inc).

Table LI. Selected Microetchants for Titanium Alloys

Name/Type	Application	Composition	Conditions
Keller's reagent (Microetchant) (89)	Keller's reagent; for etching Ti alloys	1 ml HF, 3 ml HCl, 5 ml HNO ₃ , 190 ml DI water	Immerse sample 3-10 seconds
Kroll's reagent (Microetchant) (90)	Kroll's reagent for Ti alloys; very good etch	1-3 ml HF, 2-6 ml HNO ₃ , 100 ml DI water	Swab sample 3-10 seconds or immerse sample 10-30 seconds
R-etch (Microetchant) (89)	R-etch, for Ti-Al and Ti-Al-Zr alloys	25 ml HF, 18.5 gm benzalkonium chloride, 35 ml methanol, 40 ml glycerin	Swab sample up to 20 seconds, or immerse sample up to 90 seconds with agitation
Smith (Microetchant) (91)	For etching beta-Ti alloys heat-treated below beta transus; reveals grain boundaries	60 ml propionic acid, 20 ml HNO ₃ , 10 ml HF	Swab sample
Gallaugh (Microetchant) (91)	For etching Ti-Cu alloys	60 ml lactic acid, 20 ml HNO ₃ , 10 ml HF, 10 ml DI water	Immerse sample 3-10 seconds

Table LI (conti.). Selected Microetchants for Titanium Alloys

Name/Type	Application	Composition	Conditions
Gurevich (Microetchant) (91)	For Ti alloys; reveals beta phase	10 ml HF, 10 ml HNO ₃ , 10 ml H ₂ SO ₄ , 1 gm succinic acid	Heat to 40-50°C, immerse sample 20-180 seconds
Ogden and Holden (Microetchant) (91)	For etching Ti alloys; differentiates phases in alpha-beta alloys	1-4 ml HCl, 1 ml H ₂ SO ₄ , 95-98 ml DI water	Use boiling, 3-10 minutes
Weck's tint etch (Microetchant) (91)	Weck's tint etch for Ti and alloys; alpha grains and twins colored according to orientation	5 gm ammonium bifluoride, 100 ml DI water	Immerse sample at 20°C for a few seconds
Beraha's tint etch (Microetchant) (91)	For as-cast Ti alloys; colors alpha matrix blue or green, TiC yellow or dark brown	2-3 gm sodium molybdate, 5 ml HCl, 1-2 gm ammonium bifluoride, 100 ml DI water	Immerse sample at 20°C until surface is colored
Stain removal (Microetchant) (92)	Removes etchant stains for most titanium and titanium alloys	1 ml HF, 2 ml HNO ₃ , 50 ml H ₂ O ₂ , 47 ml DI water	Immerse sample for several seconds

Table LII. Selected Macroetchants for Titanium Alloys

Name/Type	Application	Composition	Conditions
Titanium (Macroetchant) (93)	Macroetchant for Ti and Ti-base alloys; Ti- Al-Mo alloys	50 ml DI water, 40 ml HNO ₃ , 10 ml HF (40%)	Etch 5-8 minutes at 60-80°C (content of HF may be decreased)
Titanium (Macroetchant) (94)	Reveals flow lines and defects	15 ml HNO ₃ , 10 ml HF, 75 ml DI water	Etch for about 2 minutes
Titanium (Macroetchant) (95)	General purpose etch for alpha + beta alloys	50 ml HCl, 50 ml DI water	Etch for about 2 minutes
Titanium (Macroetchant) (93)	Macroetchant for Ti	30 ml DI water, 10 ml HF (40%), 60 ml H ₂ O ₂ (30%)	Etch by swabbing until desired contrast is obtained
Titanium welds (Macroetchant) (96)	Macroetchant for welded seams	200 ml DI water, 10 ml HF (40%), 10 gm Fe(NO ₃) ₃ ·9H ₂ O	Etch seconds to minutes at 50-60°C